

Port Security Implementation Using GNS3 & kali

1. Introduction

Due to the increasing risk of network access attacks such as MAC Spoofing, it has become essential to implement Layer 2 security mechanisms on network switches. Port Security is one of the most effective methods to prevent unauthorized devices from accessing the network by limiting and controlling MAC addresses per port.

In this project, a simulated enterprise network was built using GNS3, containing multiple VLANs for different departments, a FortiGate firewall for Layer 3 security, and end-user devices. The main objective was to implement Port Security on access ports to prevent unauthorized hosts from accessing the network.

2. Project Topology

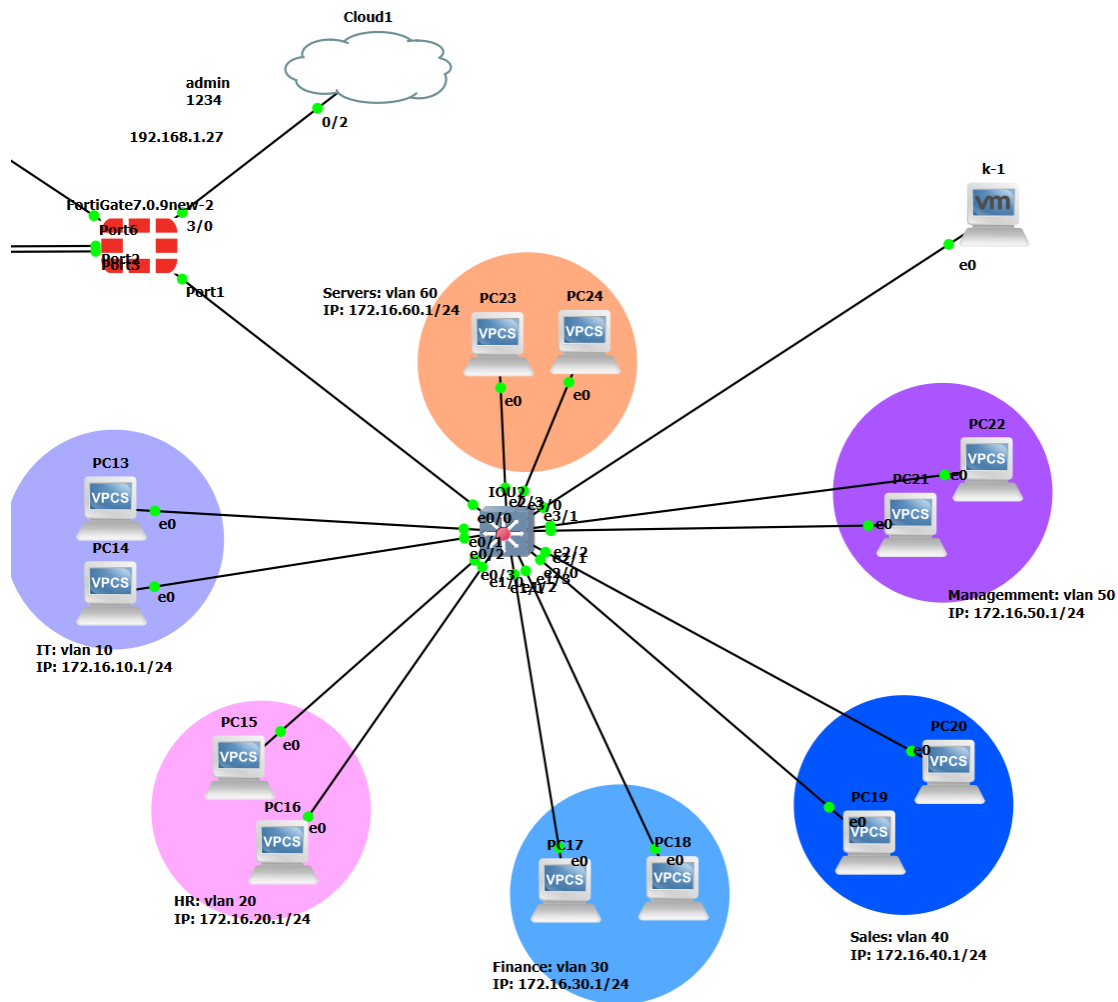
Components Used:

- GNS3 Simulator
- Layer 2 Cisco Switches (IOU/IOS)
- FortiGate Firewall
- End-user PCs distributed on multiple VLANs
- Kali Linux machine to simulate attacker

VLAN Structure:

VLAN	Department	Network
10	IT	172.16.10.0/24
20	HR	172.16.20.0/24
30	Finance	172.16.30.0/24
40	Sales	172.16.40.0/24
50	Manage	172.16.50.0/24
60	Servers	172.16.60.0/24

Each VLAN has dedicated hosts connected through different switch access ports.



3. Problem Statement

A malicious user attempts to gain unauthorized access to the network by spoofing the MAC address of a legitimate host. This allows the attacker to bypass traditional IP-based security and communicate as if they were a trusted device.

Without Port Security, the attacker can:

- Change their MAC address to match a legitimate device
- Join the VLAN
- Communicate with the default gateway
- Access internal resources

To prevent this, Port Security must be implemented on access ports.

4. Attack Scenario (MAC Spoofing Attack)

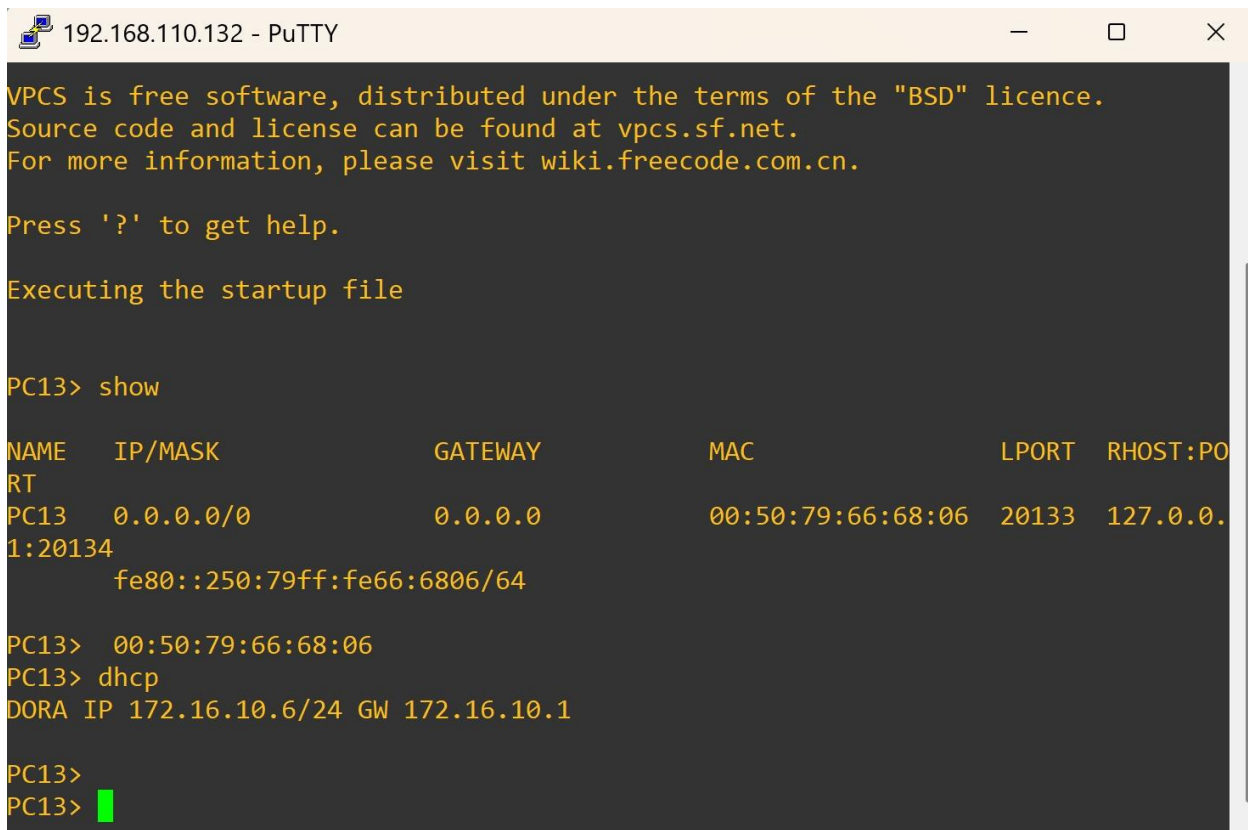
The attacker (Kali Linux) connects to the network and changes their MAC address to match that of a legitimate PC.

Steps performed by the attacker:

```
ip link show eth0
sudo ip link set dev eth0 down
sudo ip link set dev eth0 address < MAC Address>
sudo ip link set dev eth0 up
sudo ip addr add 172.16.10.50/24 dev eth0
sudo ip route add default via 172.16.10.1
ping 172.16.10.1
```

Result before Port Security:

The ping request succeeds → network access is granted → the attack is successful.



The screenshot shows a PuTTY terminal window titled "192.168.110.132 - PuTTY". The terminal output shows the VPCS startup sequence, including a network configuration table. The user enters the command "show", and the terminal displays the following table:

NAME	IP/MASK	GATEWAY	MAC	LPORT	RHOST:PORT
PC13	0.0.0.0/0	0.0.0.0	00:50:79:66:68:06	20133	127.0.0.1:20134

Below the table, the terminal shows the user entering the command "00:50:79:66:68:06" and "dhcp", followed by the output "DORA IP 172.16.10.6/24 GW 172.16.10.1". The terminal ends with a green cursor on a new line.

```
File Edit View VM Tabs Help
library
Type here to search
My Computer
gns3
Windows Server 2022
gns3
Windows Server 2022
kali@kali: ~
$ ip link show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
link/ether 00:0c:29:97:78:5c brd ff:ff:ff:ff:ff:ff
(kali@kali)-[~]
$ sudo ip link set dev eth0 down
[sudo] password for kali:
(kali@kali)-[~]
$ sudo ip link set dev eth0 address 00:50:79:66:68:06
(kali@kali)-[~]
$ sudo ip link set dev eth0 up
(kali@kali)-[~]
$ ip link show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
link/ether 00:50:79:66:68:06 brd ff:ff:ff:ff:ff:ff permaddr 00:0c:29:97:78:5c
(kali@kali)-[~]
$ sudo ip addr add 172.16.10.50/24 dev eth0
(kali@kali)-[~]
$ sudo ip route add default via 172.16.10.1
(kali@kali)-[~]
$ ping 172.16.10.1
PING 172.16.10.1 (172.16.10.1) 56(84) bytes of data:
64 bytes from 172.16.10.1: icmp_seq=1 ttl=255 time=9.61 ms
64 bytes from 172.16.10.1: icmp_seq=2 ttl=255 time=9.68 ms
64 bytes from 172.16.10.1: icmp_seq=3 ttl=255 time=6.18 ms
64 bytes from 172.16.10.1: icmp_seq=4 ttl=255 time=3.64 ms
64 bytes from 172.16.10.1: icmp_seq=5 ttl=255 time=12.2 ms
64 bytes from 172.16.10.1: icmp_seq=6 ttl=255 time=8.84 ms
64 bytes from 172.16.10.1: icmp_seq=7 ttl=255 time=8.99 ms
64 bytes from 172.16.10.1: icmp_seq=8 ttl=255 time=10.8 ms
64 bytes from 172.16.10.1: icmp_seq=9 ttl=255 time=5.14 ms
64 bytes from 172.16.10.1: icmp_seq=10 ttl=255 time=10.5 ms
64 bytes from 172.16.10.1: icmp_seq=11 ttl=255 time=8.11 ms
^C
  172.16.10.1 ping statistics:
  11 packets transmitted, 11 received, 0% packet loss, time 10018ms
 rtt min/avg/max/mdev = 3.641/8.518/12.182/2.456 ms
(kali@kali)-[~]
$ ss
```

5. Port Security Configuration

Port Security was applied to all access ports connected to end-user devices. This prevents any unauthorized MAC address from connecting to the network.

Configuration Example:

```
enable
conf t
interface e3/1
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
```

end

Explanation:

Command	Function
switchport port-security	enables Port Security
maximum 1	only one MAC allowed
sticky	auto-learns legitimate MAC addresses
violation shutdown	port shuts down if violation occurs

6. Results After Port Security

When the attacker attempted the MAC spoofing attack again, the switch detected a violation and shut down the port automatically.

Output Example:

```
Port Status          : Secure-shutdown
SecurityViolation Count : 1
```

Result:

- The attacker was disconnected
- Network remained protected
- Unauthorized access was blocked successfully

```
IOU2#show port-security int e3/1
Port Security           : Enabled
Port Status             : Secure-shutdown
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type               : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 0
Configured MAC Addresses : 0
Sticky MAC Addresses    : 0
Last Source Address:Vlan : 0050.56c0.0002:10
Security Violation Count : 1
```

```
IOU2#
```

```

IOU2(config)#int e2/1
IOU2(config-if)#switchport mode access
IOU2(config-if)#switchport port-security
IOU2(config-if)#switchport port-security maximum 1
IOU2(config-if)#switchport port-security mac-address sticky
IOU2(config-if)#switchport port-security violation shutdown
IOU2(config-if)#do wr
Building configuration...
Compressed configuration from 4013 bytes to 1732 bytes[OK]
IOU2(config-if)#exit
IOU2(config)#int e2/2
IOU2(config-if)#switchport mode access
IOU2(config-if)#switchport port-security
IOU2(config-if)#switchport port-security maximum 1
IOU2(config-if)#switchport port-security mac-address sticky
IOU2(config-if)#switchport port-security violation shutdown
IOU2(config-if)#do wr
Building configuration...
Compressed configuration from 4013 bytes to 1732 bytes[OK]
IOU2(config-if)#exit
IOU2(config)#int e2/3
IOU2(config-if)#switchport mode access
IOU2(config-if)#switchport port-security
IOU2(config-if)#switchport port-security maximum 1
IOU2(config-if)#switchport port-security mac-address sticky
IOU2(config-if)#switchport port-security violation shutdown
IOU2(config-if)#do wr
Building configuration...
Compressed configuration from 4013 bytes to 1732 bytes[OK]
IOU2(config-if)#end
IOU2#
*Dec 5 19:25:21.163: %SYS-5-CONFIG_I: Configured from console by console
IOU2#show port-security
Secure Port  MaxSecureAddr  CurrentAddr  SecurityViolation  Security Action
      (Count)         (Count)         (Count)
-----
Et0/1          1             0             0             Shutdown
Et0/2          1             0             0             Shutdown
Et0/3          1             0             0             Shutdown
Et1/0          1             0             0             Shutdown
Et1/1          1             0             0             Shutdown
Et1/2          1             0             0             Shutdown
Et1/3          1             0             0             Shutdown
Et2/0          1             0             0             Shutdown
Et2/1          1             0             0             Shutdown
Et2/2          1             0             0             Shutdown
Et2/3          1             0             0             Shutdown
Et3/1          1             0             1             Shutdown
-----
Total Addresses in System (excluding one mac per port)    : 0
Max Addresses limit in System (excluding one mac per port) : 4096
IOU2#

```

7. Conclusion

Port Security proved to be an effective method to secure the enterprise network and prevent MAC-based spoofing attacks. By limiting the number of MAC addresses allowed per port and shutting down the port upon violation, unauthorized access was fully eliminated.

This project demonstrates the importance of implementing Layer 2 security alongside firewalls and VLAN segmentation to ensure full network protection.